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END Semester Examination Jun 2025

Name of the Course: B.Tech CIVIL ENGINEERING

Name of the Paper: Transportation Engineering-I

Time: 3 Hours

Semester: VI

Paper Code: TCE 605

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1		(20marks)	
(a)	What are the different modes of transportation? Explain the specific functions of each of them.		CO1
(b)	The radius of horizontal circular curve is 100 m. The design speed is 50 kmph and the design coefficient of lateral friction is 0.15. Calculate the superelevation required if full lateral friction is assumed to develop.		
(c)	Calculate the safe stopping sight distance on a level road stretch for design speed of 50 kmph for (i) two-way traffic on a two lane road (ii) two-way traffic on a single lane road.		
Q2		(20marks)	
(a)	Discuss the principles, applications and limitations of Direct shear test.		CO2
(b)	List different types of cutbacks. When are these used?		
(c)	List and explain the purpose of materials used for construction of cement concrete pavement.		
Q3		(20marks)	
(a)	What is the object of traffic volume study?		CO3
(b)	Estimate the theoretical capacity of a traffic lane with one way traffic flow at a stream speed of 40 kmph. Assume the average space gap between vehicles to follow the relation $S_g = 0.278 V t$, where V is the stream speed in kmph, t is the average reaction time = 0.7 sec ; assume average length of vehicles = 5.0 m.		
(c)	Draw neat sketch of at least 2 regulatory signs and warning signs.		
Q4		(20marks)	
(a)	Draw a neat sketch of components of a typical flexible pavement.		CO4
(b)	Briefly outline the advantages and limitations of rigid pavement.		
(c)	Total weight of a dual wheel load assembly is 12,000 kg. The centre to centre spacing and clear space in the dual wheels are 30 cm and 10 cm respectively. Calculate the ESWL for pavement thickness 40 cm.		
Q5		(20marks)	
(a)	Length of runway under standard condition is 1620 m. the airport site has an elevation of 230 m. if the reference temperature is 32.94°C & runway is to be constructed with effective gradient of 0.2% then determine the corrected runway length.		CO5
(b)	What is a Wind rose diagram explain Briefly.		
(c)	Explain airport and their types. What are the main consideration in selection of site for an airport planning?		

